



ROWAD
MODERN ENGINEERING

Variation Order Process
P-OP-02

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	Variation Order Process		
	Effective Date: April, 2021	Rev. No.: 00	Doc. No.: P-OP-02

1. OBJECTIVES

The process defines the steps taken for estimation study till the variation submission and approval.

2. STAKEHOLDER

- Site
- Project Manager
- Technical Office Manager
- Project Control Engineer
- Operation Department
- Projects Manager
- TM
- Contract Department
- Commercial Department
- Tender Department
- Top Management
- Operation Department
- COO

3. LIABLE STAKEHOLDER

- COO
- Tender Department
- Contract Department
- OPM
- PMs
- TM
- PM
- TOM
- PCE

4. TERMS AND DEFINITION

- COO: Chief Operations Officer.
- OPM: Operation Manager
- PMS: Projects Manager
- TM: Technical Affairs Manager
- PM: Project Manager
- TOM: Technical Office Manager
- PCE: Project Control Engineer
- VO: Variation Order
- EOT: Estimation of Time

	Variation Order Process		
	Effective Date: April, 2021	Rev. No.: 00	Doc. No.: P-OP-02

- QS: Quantity Surveyor
- HO: Head Office
- SWI: Site Work Instruction
- RFI: Request for Information
- IFC: Issued for Construction
- MS: Method Statement
- RFV: Request for Variation
- NOV: Notice of Variation

5. RELATED DOCUMENTED INFORMATION

- Variation Order Initial Internal Form (F-P-OP-02-01)
- Variation Order Final Internal Form (F-P-OP-02-02)
- RFV (F-P-OP-02-03)
- Price Breakdown (F-P-OP-02-04)
- VO log (F-P-OP-02-05)
- VO Monthly Status Report (summary) (F-P-OP-02-06)

6. PROCESS DETAILS

6.1 PROCESS INPUT

- Source of Variation
- Supporting Documents

6.2 Process Operations

Sr.	Process Cycle description	Responsibility	Form
<u>SOURCE OF VARIATION</u>			
1	The Variation Order is an alternation to the scope of works in the form of an addition, substitution, or omission from the original scope of works, the Variation is initiated by several means for example: – SWI/letter by client. – Verbal Instruction by engineer/client		

	Variation Order Process		
	Effective Date: April, 2021	Rev. No.: 00	Doc. No.: P-OP-02

	– Changes related to IFC drawings. – RFI/MS replies. – Minutes of Meeting – Comments in shop drawings, Material Submittals, etc.		
2	<u>Variation Identification</u> TOM to identify the effected items “Technical Issue”.	TOM	
<u>NOV-Case1: The Contract includes a Change Management Plan</u>			
3	– TOM prepares the Client Notification Form. – TOM submit the technical issue to Contract dept within 7days from receipt of the source of Variation. – In case the cost study required revised shop drawings, and/or submittals approval, which will exceed the duration of detailed breakdown submission in the Contract. Then, PM/TOM shall notify the Contract dept. with the case, to be mentioned in notification and/or in a separate letter. – TOM submits the technical issue with rough estimated QTY to PCE within 7days from receipt of the source of Variation. (Copy to Contract Department).	TOM	
4	If preliminary time impact required by Contract, PCE study the time impact/Activity duration, and submit it to Contract Department within 5 days. (Copy to TOM) If not required, it will be studied and submitted with VO breakdown. (step 14 & 15).	PCE	
5	Contract Department reviews the Merit of VO and verify the contractual case with Contract clauses’ references.	Contract dept	

 ROWAD MODERN ENGINEERING	Variation Order Process		
Effective Date: April, 2021	Rev. No.: 00	Doc. No.: P-OP-02	

	In case VO is merited; the contract department submits the Contractual/Legal Merits to PM/TOM within 5 days from receiving of technical issue.		
6	<u>Official Submission of VO Notification:</u>		
	– PM/TOM submits the Notification form to client, within the time for VO Notification as per Contract Conditions. – In case of no determined time for VO Notification in contract; PM/TOM to submit the Notification form to client, within 14 days of Source of variation. In case of no form provided by the client, a letter to be issued by contract dept., refer to step 5.	PM/TOM	
<u>NOV-Case2: The Contract doesn't include a Change Management Plan</u>			
7	– TOM submit the Technical issue to contract dept. within 7days from receiving of the source of Variation. – In case the cost study required revised shop drawings, and/or submittals approval, which will exceed the duration of detailed breakdown submission in the Contract. Then, PM/TOM shall notify the Contract dept. with the case, to be mentioned in notification and/or in a separate letter. – TOM submits the Technical issue with rough estimated QTY to PCE within 7days from receiving of the source of Variation. (Copy to Contract Department).	TOM	
8	If preliminary time impact required by Contract, PCE study the time impact/Activity duration, and submit it to Contract Department within 5 days. (Copy to TOM) If not required, it will be studied and submitted with VO breakdown. (step 14 & 15)	PCE	

	Variation Order Process		
	Effective Date: April, 2021	Rev. No.: 00	Doc. No.: P-OP-02

9	Contract Department reviews the Merit of VO and verify the contractual case with Contract clauses' references. In case VO is merited; the contract department submits a Notification Letter to PM/TOM within 5 days from receipt of technical issue.	Contract dept	
10	<u>Official Submission to VO Notification:</u>		
	– PM/TOM submit the Notification via letter to client, within the time for VO Notification as per contract conditions. – In case of no determined time for VO Notification in contract; PM/TOM to submit the Notification form to client, within 14 day of Source of variation. – TOM to submit a copy of submitted letter to: • Operation department. • Contract department. • Project Control Department	PM/TOM	
<u>Variation Breakdown Study</u>			
11	<u>Cost of Variation</u> TOM Prepares the Cost Breakdown, along with QS.	TOM	
12	TOM prepares the <u>initial</u> internal form.	TOM	F-P-OP-02-01
13	<u>INTERNAL APPROVAL CYCLE</u> - <u>CAT-1: (up to 500K EGP).</u> 1. TOM prepares the VO. <u>initial</u> internal form. 2. TOM submits the VO. internal form to PCE & PM for approval. 3. TOM to send a copy of approved VO. Internal Form to:		F-P-OP-02-01

 ROWAD MODERN ENGINEERING	Variation Order Process		
Effective Date: April, 2021	Rev. No.: 00	Doc. No.: P-OP-02	

	Operation dept Procurement dept Project Control dept - <u>CAT-2 (500K to 5M EGP):</u> 1. TOM prepares the VO. <u>initial</u> internal form. 2. TOM submits the VO. <u>initial</u> internal form to PCE & PM for approval. 3. TOM to submit the approved VO. <u>initial</u> internal Form along with all supported documents to Operation Department for TM and PMs' approvals. 4. Operation department to send a copy of approved VO. Internal Form to: TOM, PCE and PM. Procurement department. Project Control Department. - <u>CAT-3 (5M to 10M EGP): -</u> 1. Follow CAT-2 process, plus OPM approval. - <u>CAT-4 (More than 10M EGP):</u> 1. Follow CAT2 (Item 1 to 4). 2. Operation department Submit the VO. internal form to Tender department for review and approval, with a copy to Commercial department for information. 3. After Tender approval, Operation department to submit the VO. internal form to OPM for approval. 4. After OPM approval and signature, Operation department to send a copy of approved VO. Internal Form to: 5. TOM, PCE and PM. Procurement department. Project Control Department COO.		
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	Variation Order Process		
Effective Date: April, 2021	Rev. No.: 00	Doc. No.: P-OP-02	

<u>VO Breakdown- Official Submitting to Client</u> (After VO. internal Form final approval as per each Category approval cycle)			
14	<u>Case1: The Contract includes a Change Management Plan</u>		
	- TOM submit to the Contract dept. a draft of VO breakdown – 2 days before submission officially to Client for comments (if any). - TOM submit the VO breakdown as per the change plan form and within time as per contract. - In case time for submission not determined in contract; TOM submits the breakdown to Client/Engineer for approval within 28 days from VO Official notification. - TOM to submit a copy of submitted RFV to: • Operation Department. • Contract Department. • Project Control Department	TOM	As per Contract
15	<u>Case2: The Contract doesn't include a Change Management Plan</u>		
	- TOM submit to the Contract dept. a draft of VO breakdown – 2 days before submission officially to Client for comments (if any). - TOM submit RFV to Client/Engineer within time as per contract conditions. - In case time for submission not determined in contract; TOM Submit RFV to Client/Engineer for approval within 28 days from VO Official notification. - TOM to submit a copy of submitted RFV to: • Operation department • Contract department • Project Control Department	TOM	F-P-OP-02-03 F-P-OP-02-04

	Variation Order Process		
	Effective Date: April, 2021	Rev. No.: 00	Doc. No.: P-OP-02

16	<u>FOLLOW UP</u> - Follow up Potential/under study VO's. - Follow up Submission of VO. - Follow up Engineer/Client approval. - Submit the VO log with summary to HO along with approved VO before 15th of each month.	TOM	F-P-OP-02-05 F-P-OP-02-06
	17		Revise VO. <u>Final</u> Internal Form according to client approved VO (to reflect the deviation on Overhead and Profit)
		TOM	F-P-OP-02-02

6.3 PROCESS OUTPUT

- VO. Internal Form by TOM.
- RFV by TOM.
- VO log.

7. PROCESS RISK ASSESMENT

Included in the risk register form No. F-P-QD-05-01 "Risk identification and assessment" maintained by the HOD, along with the action plan to address risks and opportunities".

8. PROCESS CONTROL

- Review and approval of PM for VO Category-1
- Review and approval of TM/PMs for VO Category-2
- Review and approval of OPM for VO Category-3
- Review and approval of Tender, commercial department for VO Category-4, and submit a copy to COO for information

9. PERFORMANCE MEASURES

- Department manager shall measure the followings:
 - Annual amounted hit ratio (value of Approved VO (Direct Cost) vs value of Studied VO (Direct Cost) shall be not exceeding 10% deviation.

	Variation Order Process		
	Effective Date: April, 2021	Rev. No.: 00	Doc. No.: P-OP-02

- Average time to submit the VO to Engineer/Client shall be not exceeding 28days or as per Contract condition.

10. DOCUMENT CHANGE HISTORY

No.	Revision Level	Description of Changes	Effective Date
01	00	New Issue	Apr 2021

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